the cloud. Want to dive into more details? Check out https://aws.amazon.com/compliance/shared-responsibility-model/.

5.2 Keeping the operating system up-to-date

Not a week goes by without the release of an important update to fix security vulnerabilities in some piece of software or another. Sometimes the kernel is affected or libraries, like OpenSSL. Other times, it's affecting an environment like Java, Apache, and PHP, or an application like WordPress. If a security update is released, you must install it quickly, because the exploit may have already been released, or because unscrupulous people could look at the source code to reconstruct the vulnerability. You should have a working plan for how to apply updates to all running virtual machines as quickly as possible.

Amazon Linux 2 installs critical or important security updates automatically on startup while `cloud-init` is running. We highly recommend you install all the other updates as well. The following options are available:

- *Install all updates at the end of the boot process*—Include `yum -y update` in your user-data script. `yum` is the package manager used by Amazon Linux 2.
- *Install security updates at the end of the boot process only*—Include the `yum -y --security update` in your user-data script.
- Use the *AWS Systems Manager Patch Manager*—Install updates based on a patch baseline.